

ChoreoSurf: Scalable Surface System with 8-DOF SMA Actuators

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Figure 1: A combination example of the ChoreoSurf system

Abstract

The ChoreoSurf system is a scalable surface system with a shape-memory alloy actuator that can bend in eight directions. This system can mount actuators on the surface layers of various three-dimensional shapes. Applications include a tabletop system, interactive wall, tentacles tower, kinetic dress, and kinetic wig.

CCS Concepts

• Applied computing → Media arts; • Hardware → Sensors and actuators.

Keywords

Shape Changing Interface, Soft Actuator, Shape-Memory Alloy

ACM Reference Format:

Akira Nakayasu. 2024. ChoreoSurf: Scalable Surface System with 8-DOF SMA Actuators. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697869>

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697869>

1 Introduction

This research is a report on a scalable surface system with a shape-memory alloy (SMA) actuator as tentacles. We have conducted several experiments and produced interactive artworks [Nakayasu 2020] inspired by the waving tentacles of sea anemones. In this project, we have successfully developed an eight-degree-of-freedom (8-DOF) actuator and detachable actuator unit.

There are several works like this project that utilize a large number of biological moving components. Daan Roosegaarde [Roosegaarde 2012] and Philip Beesley [Beesley 2009] are excellent artists who produce biological kinetic artwork. In particular, Philip's work is designed in an organic and biological manner, not only in terms of the movement of the actuators but also as a whole. On the other hand, this project aims to express a feeling of grotesqueness and vitality through the fusion of the circuit board and bared wiring with the minimalistic actuator.

2 Improvements in the Actuator Unit

2.1 A Detachable Actuator Unit

In this project, we have developed a detachable actuator unit with a height of 15 mm (Fig. 3). It will be possible to mount actuators on the surface layers of various three-dimensional shapes. Furthermore, using multiple types of base circuit boards, the SMA actuators can be placed in any position and used to create various shapes of artwork (Fig. 1 and 2).

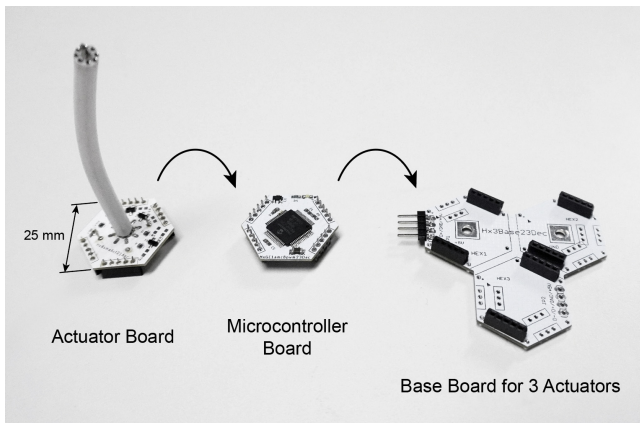


Figure 2: Structure of an actuator unit.

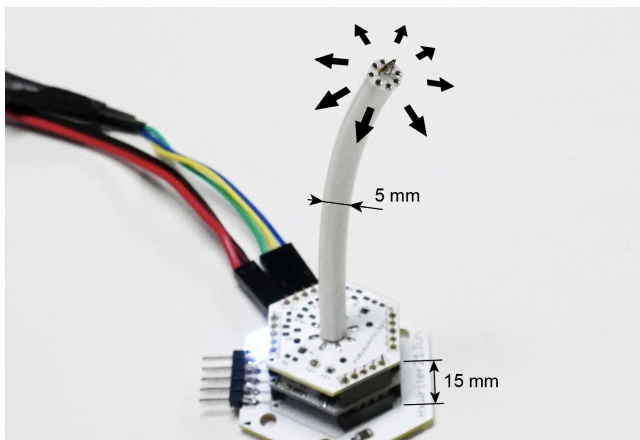


Figure 3: An 8-DOF SMA actuator.

2.2 An 8-DOF SMA Actuator

The SMA actuator (Fig.3) is composed of a silicon tube (5 mm in diameter), eight BioMetal Fibers (a thin SMA wire developed by TOKI Corporation). The previous project utilized the six directional bending actuator. In this project, the silicon tubes has been redesigned to bend in eight directions.

3 Future Applications

Although this system is still in the early stage of development, possible applications include an interactive tabletop artwork (Fig. 4) that takes advantage of the scalability with multiple types of circuit boards, and a tentacles tower or interactive wall with thousands of tentacles (Fig. 5). Furthermore, implementation in dresses or wigs will expand the possibilities for fashion and entertainment applications (Fig. 6).

Acknowledgments

This work was supported by JSPS KAKENHI 22K02846.

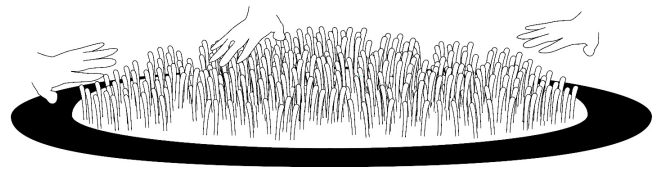


Figure 4: A concept illustration of an interactive tabletop artwork.



Figure 5: A concept illustration of a tentacles tower and interactive wall.

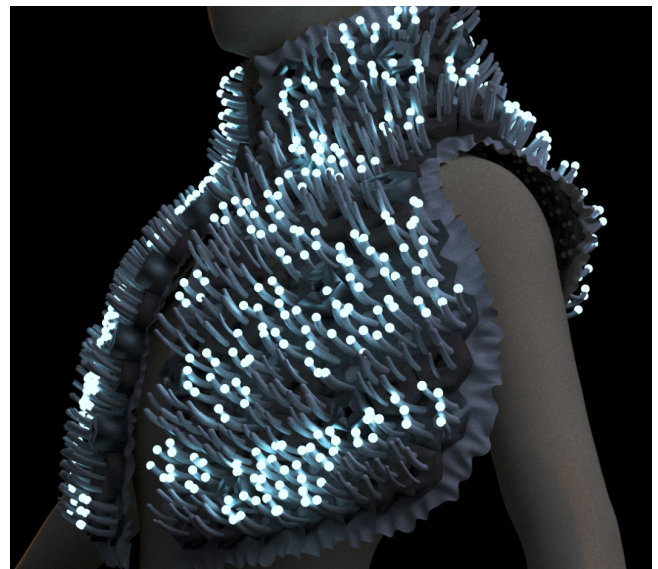


Figure 6: A concept illustration of a kinetic dress. (c) Masaharu Ono

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